



CHRIST
(DEEMED TO BE UNIVERSITY)
BANGALORE · INDIA

AGRIMITRA : AI-Powered Smart Farming Assistant

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MISSION

CHRIST is a nurturing ground for an individual's holistic development to make effective contribution to the society in a dynamic environment

VISION

Excellence and Service

CORE VALUES

Faith in God | Moral Uprightness
Love of Fellow Beings
Social Responsibility | Pursuit of Excellence

Agenda

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- Problem Statement
- Objectives
- Hardware & Software Requirements
- AI & ML Components
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Abstract

AgriMitra is an AI-powered smart farming assistant designed to provide personalized agricultural support to farmers. The platform integrates IoT-based environmental monitoring, weather data, intelligent irrigation recommendations, machine learning-based crop recommendation and rotation planning, crop price prediction, and an AI-powered agricultural assistant. Real-time farm context such as soil moisture, temperature, humidity, crop growth stage, weather conditions, and irrigation status is used to generate personalized recommendations. The platform also supports multilingual AI interaction through voice input and output, making agricultural guidance more accessible to farmers.

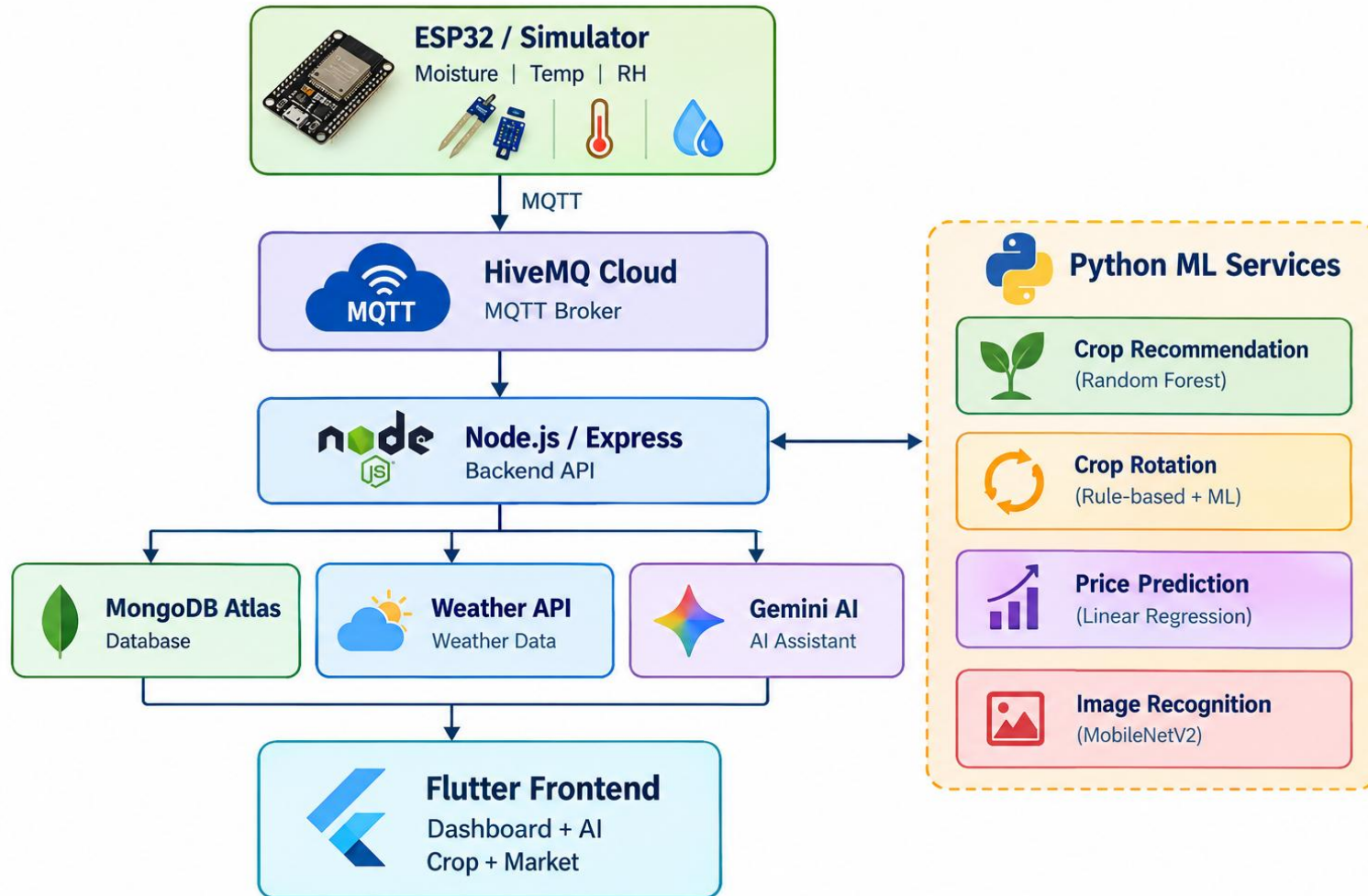
Problem Statement

- Farmers often lack intelligent tools to plan suitable crop rotations and farming activities based on soil conditions, previous crops, and seasonal factors.
- Farmers face difficulty in identifying crops from images and obtaining reliable estimates of their current and future market prices.
- Farmers need an accessible system for real-time farm monitoring and personalized agricultural guidance through simple voice-based interaction.

Objectives

- To develop a machine learning and rule-based crop rotation and scheduling system that recommends suitable crops and farming activities based on soil conditions, previous crop history, crop suitability, and seasonal factors. (phase1)
- To develop an image-based crop price prediction system that uses MobileNetV2 for crop identification and machine learning-based analysis of historical market data to estimate crop prices and future price trends. (phase 1)
- To develop an IoT-based farm monitoring prototype integrated with a multilingual AI voice assistant to provide real-time farm information and personalized agricultural guidance to users. (phase 2)

System Architecture



Hardware and Software Requirements

Component	Purpose
ESP32	IoT sensor connectivity
Capacitive Soil Moisture Sensor	Soil moisture monitoring
DHT22	Temperature & humidity
Relay Module	Planned pump switching
12V DC Pump	Proposed irrigation actuator
Water Level Sensor	Planned dry-run protection
Power Supply	Development/testing

Software Stack

Layer	Technology
Frontend	Flutter / Dart
Backend	Node.js / Express
Database	MongoDB Atlas
IoT Communication	MQTT / HiveMQ Cloud
AI	Google Gemini 2.5 Flash
Crop ML	Python / Flask / Random Forest
Price ML	Python / Flask
Image Recognition	MobileNetV2
Weather	OpenWeatherMap
Market Data	AGMARKNET / data.gov.in
Authentication	JWT + bcrypt
Voice	speech_to_text + flutter_tts

AI & ML Components

AI Component

- **Gemini 2.5 Flash**

Uses real-time farm context, weather information, crop details, and Irrigation information to provide personalized agricultural assistance.

- **Voice Assistance**

Speech-to-text and text-to-speech enable multilingual voice interaction.

ML Components

Application	Model / Algorithm
Crop Recommendation	Random Forest Classifier
Crop Rotation	Hybrid ML + Rule-based Approach
Crop Image Recognition	MobileNetV2
Price Prediction	Linear Regression / Trend Forecasting
Irrigation	Hargreaves ET_0 + K_c + Rule-based Decision Logic

Literature Review / Background Study

Paper	Methodology	Advantages	Limitations
Darshna et al. (2025) – Smart Irrigation System	Uses soil moisture sensor, temperature sensor, relay, Arduino controller, and automatic pump control.	Reduces water wastage and automates irrigation.	No IoT connectivity, no remote monitoring, no AI support.
El Mezouari et al. (2022) – Smart Irrigation System	Uses Arduino UNO, soil moisture sensor, temperature sensor, water flow sensor, relay module, and automatic irrigation	Efficient water usage and automatic monitoring.	Uses serial communication only; lacks cloud connectivity and intelligent decision-making.
Badreldeen et al. (2022) – IoT Based Smart Irrigation System	Uses ESP32, soil moisture, temperature, humidity, rain sensors, MQTT communication, mobile and web applications.	Real-time monitoring, weather integration, remote control.	No AI assistant, no multilingual voice interaction.

Proposed Solution

- The platform provides machine learning-based crop recommendations and crop rotation plans by considering soil suitability, previous crops, and relevant agricultural factors.
- The system identifies crops from images and provides crop price estimates and future price trends using market data from AGMARKNET.
- AgriMitra monitors real-time soil moisture, temperature, humidity, and weather conditions to provide farmers with up-to-date information about their farm environment.
- The system determines irrigation requirements using Hargreaves ET_0 , crop coefficient (K_c), crop growth stage, soil moisture, and rainfall probability.
- A Gemini 2.5 Flash-powered AI assistant uses real-time farm data to provide personalized agricultural guidance through multilingual text and voice interaction.

Modules

Crop Rotation & Scheduling

- Recommends suitable crop sequences based on **soil conditions and previous crops**.
- Suggests appropriate **farming activities and schedules**.
- Uses **machine learning and rule-based analysis** for crop planning.

Image-Based Crop Price Prediction

- Identifies the crop from an **uploaded image using MobileNetV2**.
- Retrieves relevant **market price data**.
- Predicts **current and future crop price trends** using historical data.

Screenshots

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Crop Rotation

done

Get AI-powered crop rotation advice based on your soil and climate.

Soil Type

▲

Red Soil

▼

Previous Crop

🔄

Groundnut

▼

🕒

27.9°C, 63% humidity - ~500mm rainfall (Estimated)

🔄

Get Recommendation

Recommendations

#1

Finger Millet

Overall Fit: 65%

Consumes nitrogen enriched by Groundnut

Model Confidence

27%

Soil Suitability

100%

Rotation Benefit

67%

- Different plant family from Groundnut (Legume) — breaks pest and disease buildup
- Nitrogen behavior: Depletor — consumes nitrogen enriched by Groundnut
- Soil suitability: Well-suited (S1, score 100/100) - Red soil well-drained semi-arid ideal for finger millet; traditional crop in red soil regions
- Needs 51-100mm water — region averages ~500mm (South Indian semi-arid average), good drainage recommended

#2

Sesame

Overall Fit: 53%

Diversifies pest exposure while matching soil moisture needs

▼

#3

Rice

Overall Fit: 40%

Consumes nitrogen enriched by Groundnut

▼

Excellence and Service

Price Prediction

Estimate current mandi prices for your produce, with optional photo detection.

Crop
tomato

State
Andhra Pradesh

District (optional)
Any district

Upload a photo of your produce (optional)
Choose Image

₹ Get Price Estimate

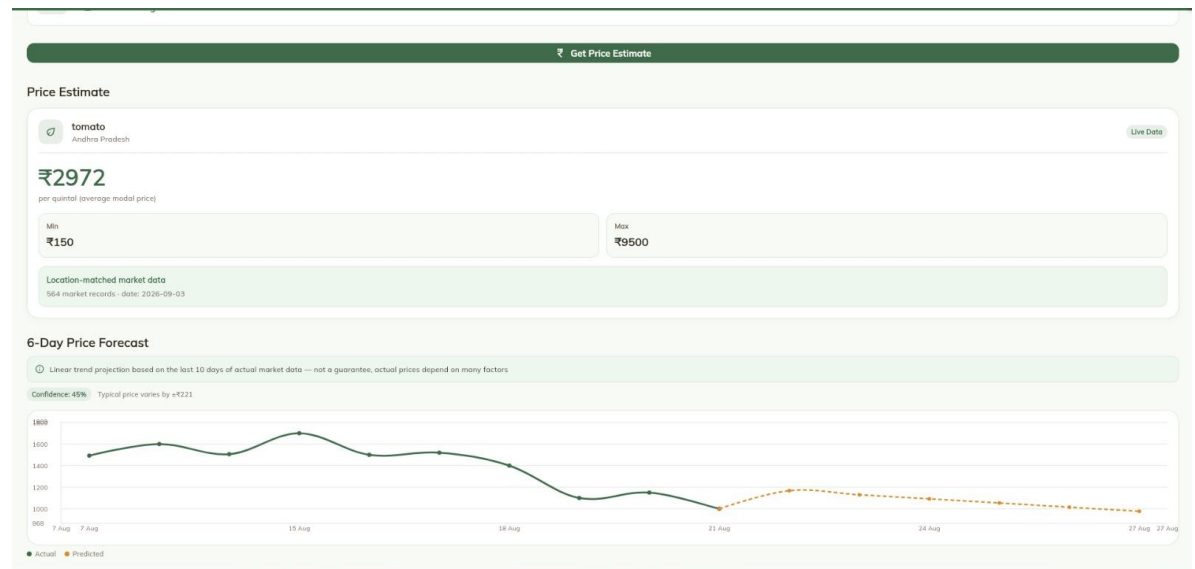
Price Estimate

tomato
Andhra Pradesh
Live Data

₹2972
per quintal (average modal price)

Min
₹150
Max
₹9500

Location-matched market data
564 market records - date: 2026-09-03



Timeline

Phase	Duration	Deliverable
Requirement analysis & literature review	Weeks 1–2	Finalized scope, reviewed prior work
Hardware setup & sensor integration	Weeks 3–4	ESP32 + sensors + relay + pump functional
Backend & database architecture	Weeks 5–6	MQTT pipeline, MongoDB, REST APIs
Irrigation logic (ET _o engine)	Weeks 7–8	Weather-grounded irrigation scheduling
Crop recommendation ML model	Week 9	Trained and served model (99.55% accuracy)
Conversational AI + voice assistant	Weeks 10–12	Grounded chatbot, multilingual voice I/O
Authentication, alerts, farm calendar	Weeks 13–14	Secure login, notification system, auto-calendar
Hardware-software full integration	Weeks 15–16	Real ESP32 replacing simulated sensor data
Testing, refinement, documentation	Weeks 17–18	End-to-end testing, report writing
Final review & viva preparation	Week 19+	Presentation, demo readiness

References

1. El Mezouari, A. E., Fazziki, A. E., and Sadgal, M., "Smart Irrigation System," *IFAC-PapersOnLine*, vol. 55, no. 10, pp. 3298–3303, 2022. DOI: 10.1016/j.ifacol.2022.10.125.
2. Badreldeen et al., "Smart irrigation system based on IoT and machine learning," *Smart Agricultural Technology*, vol. 3, 2022. DOI: 10.1016/j.ssaho.2022.100147.
3. AI4Bharat — IndicASR / IndicTTS: Open-source speech models for Indian languages, IIT Madras
4. Kaggle Crop Recommendation Dataset — N-P-K, temperature, humidity, pH, rainfall to crop label dataset